

An infinite number of tosses.

Let's keep going. We can flip 4,000 times staking a tenth of a penny. How about 400,000 times at 1/1000 of a penny? As a Platonic form, the pure Gaussian curve is principally what happens when we have an infinity of tosses per round, with each bet infinitesimally small. Do not bother trying to visualize the results, or even make sense out of them. We can no longer talk about an "infinitesimal" bet size (since we have an infinity of these, and we are in what mathematicians call a continuous framework). The good news is that there is a substitute.

We have moved from a simple bet to something completely abstract. We have moved from observations into the realm of mathematics. In mathematics things have a purity to them.

Now, something completely abstract is not supposed to exist, so *please do not even make an attempt to understand [Figure 10](#)*. Just be aware of its use. Think of it as a thermometer: you are not supposed to understand what the temperature *means* in order to talk about it. You just need to know the correspondence between temperature and comfort (or some other empirical consideration). Sixty degrees corresponds to pleasant weather; ten below is not something to look forward to. You don't necessarily care about the actual speed of the collisions among particles that more technically explains temperature. Degrees are, in a way, a means for your mind to translate some external phenomena into a number. Likewise, the Gaussian bell curve is set so that 68.2 percent of the observations fall between minus one and plus one standard deviations away from the average. I repeat: do not even try to understand whether *standard deviation* is *average deviation*—it is not, and a large (too large) number of people using the word *standard deviation* do not understand this point. Standard deviation is just a number that you scale things to, a matter of mere correspondence *if phenomena were Gaussian*.

These standard deviations are often nicknamed "sigma." People also talk about "variance" (same thing: variance is the square of the sigma, i.e., of the standard deviation).

Note the symmetry in the curve. You get the same results whether the sigma is positive or negative. The odds of falling below -4 sigmas are the same as those of exceeding 4 sigmas, here 1 in 32,000 times.

As the reader can see, the main point of the Gaussian bell curve is, as I have been saying, that most observations hover around the mediocre, the mean, while the odds of a deviation decline faster and faster (exponentially) as you move away from the mean. If you need to retain one single piece of information, just remember this dramatic speed of decrease in the odds as you